Year 9 Science Test 1 Core paper

Variation and inheritance · Adaptation and extinction · Cells and the microscope · Atoms and the periodic table

Name	Class	Date	Score / 40

Instructions · Time: 40 minutes · Total: 40 marks

- You have 40 minutes. Answer all the questions.
- Write your answers in the spaces on this paper. If you need more space, use the back of the page and write the question number.
- The marks for each part are shown in brackets, for example [2]. There are 40 marks in total.
- Read each question carefully. Use the science key words in your answers.

Key words 关键词

abiotic factor	非生物因素	**biotic factor**	生物因素	**environmental variation**	环境变异
continuous variation	连续变异	**discontinuous variation**	不连续变异	**species**	物种
inherited	遗传的	**nucleus**	细胞核	**gamete**	配子
fertilisation	受精	**zygote**	受精卵	**adaptation**	适应性特征
native species	本地物种	**nature reserve**	自然保护区	**breeding programme**	繁育计划
gene bank	基因库	**cell wall**	细胞壁	**chloroplast**	叶绿体
vacuole	液泡	**cell membrane**	细胞膜	**cytoplasm**	细胞质
magnification	放大倍数	**eyepiece lens**	目镜	**objective lens**	物镜
proton	质子	**neutron**	中子	**electron**	电子
atomic number	原子序数	**mass number**	质量数	**electron configuration**	电子排布

Command words 指令词: Calculate 计算 · Complete 完成（填写） · Draw 画出 · Explain 解释 · Give 给出 · Tick 打勾 · Use 使用（运用） · Write 写出

1 Environmental variation (8 marks)

(a) Put one tick in each row to show whether the factor is abiotic (non-living) or biotic (living). **[2]**

Factor	Abiotic	Biotic
temperature		
predation (being hunted)		
amount of light		
infectious disease		

(b) A scar is caused by a cut. Complete the sentence. **[1]**

A scar is an example of _____ variation.

(c) Put one tick in each row to show whether the variation is continuous or discontinuous. **[2]**

Characteristic	Continuous	Discontinuous
height		
blood group		
hand span		
tongue rolling (can or cannot)		

(d) Some students measured the length of 40 leaves and drew a bar chart of their results. Should the bars touch each other? Tick one box and give a reason. **[2]**

- ☐ Yes, the bars should touch.
- ☐ No, there should be gaps between the bars.

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(e) What is a species? **[1]**

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2 Inherited variation

(8 marks)

(a) Complete the sentences using words from the box. Use each word once. One word is not needed. **[4]**

nucleus	**gametes**	**fertilisation**	**zygote**	**environment**

Genetic information is stored in the _____ of a cell.

A sperm cell and an egg cell are both _____.

They join together at _____.

The new cell that is made is called a _____.

(b) Tick the two characteristics that are inherited. **[2]**

- ☐ natural eye colour
- ☐ a scar
- ☐ blood group
- ☐ being able to speak Chinese

(c) Brothers and sisters have the same parents, but they are not identical. Explain why. **[2]**

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3 Adaptation and extinction

(8 marks)

(a) What is an adaptation? **[1]**

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(b) A sand cat lives in the desert. Its fur is the same colour as the sand. Explain how this helps the sand cat to survive. **[2]**

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(c) Grey squirrels were brought to the UK from the USA. Red squirrels are now rare in most of the UK. Give two reasons why grey squirrels have replaced red squirrels. **[2]**

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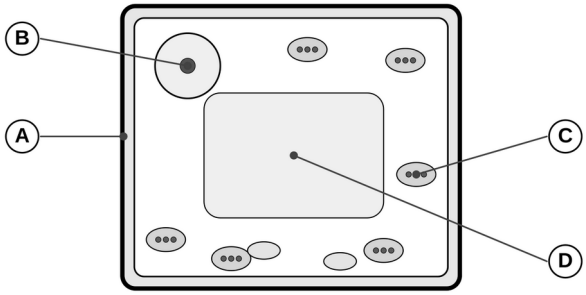
(d) Draw one line from each way of protecting a species to how it works. **[3]**

nature reserve	endangered animals are bred in captivity to increase their numbers
breeding programme	seeds, gametes or cells are stored at a low temperature
gene bank	an area where the habitat and the species living in it are protected

4 Cells and the microscope

(8 marks)

(a) The diagram shows a plant cell. Write the name of each labelled part. **[4]**



A plant cell seen through a microscope (not to scale)

A _____

B _____

C _____

D _____

(b) Which part of a cell controls what goes into and out of the cell? Tick one box. **[1]**

- ☐ cell wall
- ☐ cell membrane
- ☐ cytoplasm
- ☐ nucleus

(c) A microscope has a x10 eyepiece lens and a x4 objective lens. Calculate the total magnification. **[1]**

Total magnification = x _____

(d) Give two rules for a good scientific drawing of cells. **[2]**

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5 Atoms and the periodic table (8 marks)

(a) Complete the table about the three particles in an atom. The first row has been done for you. **[4]**

Particle	Relative charge	Where it is found
proton	+1	in the nucleus
neutron		
electron		

(b) A sodium atom has atomic number 11 and mass number 23. **[2]**

(i) How many protons does the atom have?

protons = _____

(ii) How many neutrons does the atom have?

neutrons = _____

(c) A sodium atom has 11 electrons. Complete its electron configuration. **[1]**

2, 8, _____

(d) A magnesium atom has atomic number 12. Write its electron configuration. **[1]**

_____ , _____ , _____

END OF TEST